

Univ.-Prof. Dr. Dominik Kowald

Curriculum Vitae

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Profile

Bio I am **Professor of AI-based Information Retrieval in Digital Humanities** (25%, until 2030) at the [University of Graz](#) and **head of the research area FAIR-AI** at [Know Center](#). I hold a **venia docendi** (habilitation) and a **Ph.D. with distinction** in Applied Computer Science from [TU Graz](#). In my **Ph.D. thesis**, I utilized the cognitive architecture **ACT-R** to model human word retrieval for **NLP and machine learning** tasks, such as predicting keyword usage. During my **post-doctoral research** on trustworthy AI and recommender systems, I was a recipient of the [Styrian Provincial Government](#) mobility grant for young scientists for a **research visit** at the [XAI group](#) at [Maastricht University](#). Currently, I am a **key researcher** in the international 3.7M€ [Interfaces of Agent-Centric AI](#) COMET module, as part of which I will be an invited guest scientist at the [TU Munich Institute for Ethics in AI](#) in summer 2027. I have published more than **130 papers** in interdisciplinary and AI venues, leading to invitations to renowned **international institutions**, including [Schloss Dagstuhl](#) and the [ELLIS Unit Berlin](#), as well as to external expert appointments at the [Universities of Gothenburg](#) and [Bergen](#).

Key Achievements

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|---|---|
| Research,
Mobility, &
Internationalization | <ul style="list-style-type: none">◦ Awards: Two-time winner of the MindTheGap Gender & Diversity Award (2022 & 2025) and runner-up for Styrian Research Award 2025 for my habilitation.◦ Impact: Q1 journal and A* conference publications in venues such as Nature Scientific Reports, ACM TIST, Int. Journal of Human-Computer Studies, and TheWebConf.◦ International Network & Mobility: Co-authorship network with international institutions (e.g., University of Colorado Boulder, TU Delft). Competitive mobility funding for research visit at Maastricht University in 2021. Invited guest scientist at IEAI, TU Munich in summer 2027. Regular speaker at international conferences (e.g., TheWebConf in Australia/Korea, Hypertext in Canada, ECIR in Norway). |
| Teaching,
Supervision, &
Mentoring | <ul style="list-style-type: none">◦ Qualification: Venia Docendi (habilitation) in Applied Computer Science, Advanced Teaching Certificate, and finished Gender & Diversity Training from TU Graz.◦ Interdisciplinarity: Experience teaching computer science to diverse student groups (Computational Social Systems, Digital Humanities), with favorable evaluations.◦ Innovation: Research-oriented course outlines for e.g., Data Management (~500 students), and involvement in international student co-supervisions (e.g., NTNU). |
| Management,
Leadership, &
Project Acquisition | <ul style="list-style-type: none">◦ Leadership: Head of FAIR-AI at Know Center since 2021, and appointed expert evaluator for international faculty appointments and promotions (from post-doctoral researcher to full Professor level) at the Universities of Gothenburg and Bergen.◦ International Projects: Acquisition of three FFG COMET Module grants worth 3.7M€ each, and recently, 1.4M€ Horizon Europe funding on agentic AI and LLMs.◦ Community Building: Co-organizer of international scientific events, e.g., the HyPER workshop at ACM UMAP or the HumaneLLM workshop at ACM IUI. |

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Education

- 2017–2024 **Priv.-Doz. (Habilitation), Applied Computer Science, [TU Graz](#).**
(Venia docendi) **Thesis:** [Transparency, Privacy, and Fairness in Recommender Systems](#)
External reviewers: Prof. Robin Burke ([University of Colorado Boulder](#)), Prof. Julian McAuley ([University of California San Diego](#)), Prof. Magdalini Eirinaki ([San Jose State University](#))
- 2012–2017 **Dr.techn. (Ph.D.), Computer Science, [TU Graz](#), w/ distinction.**
(Doctoral studies) **Thesis:** [Modeling Activation Processes in Human Memory to Improve Tag Recommendations](#)
Supervisors: Prof. Stefanie Lindstaedt ([IT:U, TU Graz](#)), Prof. Elisabeth Lex ([TU Graz](#))
External supervisor: Prof. Tobias Ley ([Tallinn University, DUK](#))
- 2009–2012 **Dipl.-Ing. (MSc.), Software Engineering & Management, [TU Graz](#), w/ distinction.**
(Master's studies) **Thesis:** [Combining Computer-Supported, Collaborative Learning with E-Assessment](#)
Supervisor: Assoc. Prof. Christian Gütl ([TU Graz](#))
Co-supervisor: Assoc. Prof. Mohammad Al-Smadi ([TU Graz, Qatar University](#))
- 2006–2009 **BSc., Software Engineering & Management, [TU Graz](#).**
(Bachelor's studies) **Thesis:** [Peer Assessment in Computer Science](#)
Supervisor: Assoc. Prof. Christian Gütl ([TU Graz](#))
Co-supervisor: Assoc. Prof. Mohammad Al-Smadi ([TU Graz, Qatar University](#))

Professional Experience

- 08/2027–10/2027 **Invited Guest Scientist, [TU Munich, Institute for Ethics in AI](#), Germany.**
(Guest researcher) **Research focus:** societal and ethical aspects of LLMs, machine learning, and agentic AI; multistakeholder AI funded by the [Interfaces of Agent-centric AI](#) FFG COMET module
- since 2026 **Expert Reviewer for Professorship Promotion, [University of Gothenburg](#), Sweden.**
(External expert) **Role:** International expert evaluator to conduct formal reviews for full professor promotions at the Faculty of Computer Science and Engineering.
- 2025–2030 **University Professor (25%), [University of Graz, Department of Digital Humanities](#).**
(10h/w) **Professorship:** AI-based Information Retrieval in Digital Humanities
Research focus: LLMs, Recommender Systems, and RAG-based architectures in the humanities
Teaching and supervision focus: information retrieval, LLMs, and applied computer science
Most important international collaborations: AI in digital archives with [University of Colorado Boulder](#), multistakeholder recommender systems with [NTNU](#)
- since 2025 **Expert Reviewer for Faculty Appointments, [University of Bergen](#), Norway.**
(External expert) **Role:** International expert evaluator to assess applications across various career levels, ranging from post-doctoral researcher selections to Associate Professor.
- since 2021 **Research Area Head and Lecturer, [Know Center, FAIR-AI and TU Graz, HCC](#).**
(full-time) **Mobility:** [XAI group of Prof. Nava Tintarev, Maastricht University](#), The Netherlands (2021);
since 2025: international research visit funded by the [Styrian Provincial Government](#)
(36h/w) **Research focus:** trustworthy AI; LLMs, NLP, ML, and agentic AI; fairness; semantic search
Venia docendi: scientific subject Applied Computer Science at TU Graz (2024)
Most important international collaborations: HorizonEurope [LUMEN](#) led by [Centre National de la Recherche Scientifique, France](#), HorizonEurope [NAVI](#) led by [Idener.ai, Spain](#)
- 2017–2020 **Deputy Research Area Head and Post-doctoral Researcher, [Know Center, FAIR-AI](#).**
(Full-time) **Mobility:** [Web Information Systems Group, Delft University](#), The Netherlands (2020); international research visit funded by the [Styrian Provincial Government](#) (disrupted due to COVID-19)
Research focus: sentiment analysis and stance detection applied to social media posts, AI
Most important international collaborations: Horizon2020 [AI4EU](#) led by [Thales SIX GTS France SAS](#), Horizon2020 [TRUSTS](#) led by [Leibniz University Hannover, Germany](#)
- 2012–2017 **Ph.D. Candidate and Researcher, [Know Center, FAIR-AI and TU Graz, HCC](#).**
(Full-time) **Research focus:** model human word retrieval for NLP and ML tasks, such as keyword prediction
Most important international collaborations: FP7 [Learning Layers](#) led by [Tallinn University, Estonia](#), Horizon2020 [AFEL](#) led by [University of Galway, Ireland](#)

International Scientific Event Organization & Chairing

- 2027 **IUI Workshop**, *HumaneLLM Workshop at ACM IUI'27*, Helsinki, Finland, Role: co-organizer.
- 2026 **UMAP**, *Doctoral consortium of ACM UMAP'26*, Gothenburg, Sweden, Role: mentor.
- 2026 **UMAP**, *Trust, Fairness, and User-Centered Evaluation in Personalized Systems track of ACM UMAP'26*, Gothenburg, Sweden, Role: session chair.
- 2026 **STS Session**, *Algorithmic fairness session at the 24th Conference on Critical Issues in Science, Technology and Society Studies*, Graz, Austria, Role: co-organizer.
- 2026 **CLARIAH Hackathon**, *Efficient Similarity: A Hackathon for Advanced Search in Cultural Heritage at University of Graz*, Graz, Austria, Role: co-organizer.
- 2025 **UMAP Workshop**, *Hybrid AI for Human-Centric Personalization Workshop co-located with UMAP'25*, New York City, United States, Role: co-organizer.
- 2025 **ReproAI Workshop**, *Reproducibility and AI Workshop co-located with "Yes, we are open" Weizenbaum Institute Conference*, Berlin, Germany, Role: co-organizer.
- 2025 **STS Session**, *Fairness and AI session at the 23rd Conference on Critical Issues in Science, Technology and Society Studies*, Graz, Austria, Role: co-organizer.
- 2024-2025 **CRBAM Workshop**, *Fair recommendations for cyclists workshop at 8th and 9th Cycling Research Board Annual Meeting (CRBAM)*, Zurich, Switzerland, Role: co-organizer.
- 2022 **DIH Süd Workshop**, *Digital Innovation Hub (DIH) Süd workshop on recommender systems and trustworthy AI*, Graz, Austria, Role: co-organizer.
- 2020 **BigData Summer Academy**, *Know Center Research GmbH summer academy on recommender systems*, Graz, Austria, Role: co-organizer.
- 2018 **ACM CIKM**, *RecSys track of ACM CIKM'18*, Turin, Italy, Role: session chair.
- 2017 **RSBDA Workshop**, *Second workshop on recommender systems and big data analytics (RSBDA) at i-KNOW'17*, Graz, Austria, Role: co-organizer.
- 2016 **RSBDA Workshop**, *First workshop on recommender systems and big data analytics (RSBDA) at i-KNOW'16*, Graz, Austria, Role: co-organizer.
- 2015 **i-KNOW**, *Social Computing track at i-KNOW'15*, Graz, Austria, Role: session chair.
- 2013 **i-KNOW**, *Science 2.0 track at i-KNOW'13*, Graz, Austria, Role: session chair.

Scientific Awards, Honours, & Mobility Grants

- 2026 **RecSys Journal Paper of the Year Award**, *RecSys'26*, Co-author of the awarded paper within the Women in RecSys framework with female authors Sole Pera, Julia Neidhardt, Nava Tintarev, and Özlem Özgöbek.
- 2025 **Competitive Conference Mobility Grant**, *EWAF'25*, for traveling to the European Workshop on Algorithmic Fairness in Eindhoven, The Netherlands.
- 2025 **Mind-the-gap Gender and Diversity Award**, *TU Graz Gender & Diversity department*, for the FairRecSys project on multistakeholder fairness where I have been co-PI.
- 2025 **Runner-up Styrian Research Award**, *State of Styria*, for my habilitation.
- 2024 **Outstanding Reviewer Award**, *UMAP'24*, Cagliari, Italy.
- 2023 **RecSys Journal Paper of the Year Award**, *RecSys'23*, Co-author of the awarded paper within the Women in RecSys framework with female author Elisabeth Lex.
- 2022 **Mind-the-gap Gender and Diversity Award**, *TU Graz Gender & Diversity department*, for my research on popularity bias in AI-driven multimedia recommender systems.

- 2021 **Competitive Mobility Grant for Young Scientists**, [State of Styria](#), for a research visit on explainable AI at [Maastricht University](#), The Netherlands.
- 2020 **Competitive Mobility Grant for Young Scientists**, [State of Styria](#), for a research visit on explainable AI at [TU Delft](#), The Netherlands (disrupted due to COVID-19).
- 2019 **Competitive Conference Mobility Grant**, [ESCSS'19](#), for traveling to the Third European Symposium on Computational Social Science in Zurich, Switzerland.
- 2018 **Competitive Conference Mobility Grant**, [ESCSS'18](#), for traveling to the Second European Symposium on Computational Social Science in Cologne, Germany.
- 2018 **Ph.D. Thesis Award**, [Chamber of Labor Styria](#), for my Ph.D. thesis.
- 2016 **Competitive Conference Mobility Grant**, [Hypertext'16](#), for traveling to the 27th ACM Conference on Hypertext and Social Media in Halifax, Canada.
- 2015 **Best Demo Honourable Mention**, [i-KNOW'15](#), for hotel recommendation demo.
- 2014 **Best Poster Award**, [Hypertext'14](#), Santiago, Chile, for my TagRec framework.

International Project Grants & Research Funding

- 2026–2029 **HorizonEurope**, [REPLICATE](#), 360k€ for Know Center (360k€ for FAIR-AI), *Role*: study co-lead for AI replications.
- 2026–2029 **HorizonEurope**, [EU MARitime SECurity through Autonomous Real-time Surveillance platforms \(MARSEC\)](#), 200k€ for FAIR-AI at Know Center, *Role*: task lead on AI ethics.
- 2026–2029 **HorizonEurope**, [CompreHensive EVALuation methods for GPAI models \(CHEVAL\)](#), 580k€ for FAIR-AI at Know Center, *Role*: co-work package lead on LLM evaluation.
- 2026–2029 **HorizonEurope**, [Navigational AI assistant for Virtual worlds Immersion \(NAVI\)](#), 636k€ for Know Center (506k€ for FAIR-AI), *Role*: co-work package lead on LLMs.
- 2026–2029 **FFG COMET Module**, [Interfaces of Agent-Centric AI \(IACAI\)](#), 3.7M€ for Know Center (350k€ for FAIR-AI), *Role*: key researcher for subproject on societal aspects.
- 2026–2027 **FFG BASIS**, [CareLLM](#), 400k€ for Know Center (200k€ for FAIR-AI), *Role*: key researcher for privacy and fairness in LLMs.
- 2025–2027 **HorizonEurope**, [Linked User-driven Multidisciplinary Exploration Network \(LUMEN\)](#), 415k€ for Know Center (83k€ for FAIR-AI), *Role*: key researcher for recommendations.
- 2025–2027 **Erasmus+**, [AI For a new Design Education Approach \(AIDEA\)](#), 51k€ for FAIR-AI at Know Center, *Role*: key researcher for AI methods in design.
- 2025 **FFG FemTech**, [Internship Grant](#), 8.5k€ for FAIR-AI at Know Center, *Role*: supervisor.
- 2024–2025 **FFG BASIS**, [Pro'K'ress](#), 348k€ for Know Center (174k€ for FAIR-AI), *Role*: key researcher on privacy-aware machine learning.
- 2024–2025 **OpenWebSearch Third-Party Tender**, [Trustworthy Access to Knowledge from the Indexed Web \(TILDE\)](#), 100k€ for Know Center (33k€ for FAIR-AI), *Role*: key researcher on bias mitigation in LLMs.
- 2024–2025 **FFG AI4Green**, [Strategic AI Roadmap for Mobility \(SAIROM\)](#), 50k€ for Know Center (25k€ for FAIR-AI), *Role*: key researcher on sustainability aspects.
- 2024–2025 **Styrian AI Future Fund**, [FairRecSys](#), 74k€ for TU Graz (37k€ for FAIR-AI), *Role*: work package lead on multistakeholder AI aspects.
- 2023–2026 **FFG COMET Research Center**, [Know Center - Research Center for Trustworthy AI](#), 3.4M€ for FAIR-AI at Know Center, *Role*: research area head for FAIR-AI.
- 2022–2026 **FFG COMET Module**, [Data-Driven Immersive Analytics \(DDIA\)](#), 3.7M€ for Know Center (350k€ for FAIR-AI), *Role*: key researcher for subproject on recommendations.

- 2022–2025 **FFG FemTech**, [Radreisen4All](#), 150k€ for FAIR-AI at Know Center, *Role*: key researcher in the field of multistakeholder recommender systems.
- 2022–2023 **FFG FemTech**, [Internship Grant](#), 8.5k€ for FAIR-AI at Know Center, *Role*: supervisor.
- 2020–2023 **FFG COMET Module**, [Data-Driven Artificial Intelligence \(DDAI\)](#), 3.7M€ for Know Center (700k€ for FAIR-AI), *Role*: key researcher for subproject on XAI and private AI.
- 2020–2023 **Erasmus+**, [Cogsteps](#), 130k€ for FAIR-AI at Know Center and TU Graz, *Role*: key researcher on recommender systems for matchmaking.
- 2020–2022 **H2020**, [Trusted Secure Data Sharing Space \(TRUSTS\)](#), 730k€ for Know Center (138k€ for Social Computing), *Role*: task lead on AI-based matchmaking.
- 2020–2022 **H2020**, [TRIPLE](#), 377k€ for Know Center (120k€ for Social Computing), *Role*: co-task lead on scholarly recommender systems.
- 2020–2022 **H2020**, [AI4EU](#), 147k€ for Know Center (73.5k€ for Social Computing), *Role*: co-task lead on European data for AI.
- 2019–2022 **FFG COMET Research Center**, [Know Center - Research Center for Big Data Analytics](#), 3.4M€ for Social Computing at Know Center, *Role*: deputy research area head.
- 2019–2021 **FFG BASIS**, [Automated Marketing and Loyalty System for Retailers and Stores \(Jolioo\)](#), 120k€ for Social Computing at Know Center, *Role*: researcher for recommender systems.
- 2018–2020 **FFG BASIS**, [Studo App Sales Offensive](#), 120k€ for Social Computing at Know Center, *Role*: researcher for recommender systems.
- 2018–2020 **Styrian Health Fund**, [Health-Literacy und Diversity \(HeLi-D\)](#), 75k€ for Know Center (37.5k€ for Social Computing), *Role*: work package lead on personalization.
- 2018–2020 **OpenAIRE Third-Party Tender**, [OpenAIRE Matchmaker](#), 75k€ for Know Center, *Role*: researcher for matchmaking algorithms.
- 2015–2018 **FFG Lighthouse**, [Data Market Austria \(DMA\)](#), 286k€ for Know Center (170k€ for Social Computing), *Role*: researcher for recommender systems.

Services to the International Research Community

- since 2026 **Mentor of Ph.D. students at Doctoral Consortium**, [UMAP](#).
- 2022–2023 **Journal Special Issue Editor**, [Frontiers in Big Data - Reviews in Recommender Systems](#), (co-) editor of 10 review articles in the field of recommender systems.
- since 2021 **Journal Review Editor**, [Frontiers in Big Data - Recommender Systems Section](#).
- since 2020 **Senior Program Committee (conferences)**, [IUI](#) (full paper track), [ECAI](#) (full paper track), [RecSys](#) (full paper, short paper, and reproducibility tracks), [ECIR](#) (reproducibility track), [ECTEL](#) (full paper track).
- since 2020 **Open Datasets**, [LFMUserGroups - User Groups to Study Popularity Bias in Music Recommendation](#), [FairRecSys - Datasets for Studying Popularity Bias](#).
- since 2017 **Reviewing for Journals and Books**, [DH Quarterly](#), [Transactions on Recommender Systems](#), [Journal of Human Computer Interaction](#), [TIST](#), [Frontiers in Psychology](#), [Applied Soft Computing](#), [EPJ Data Science](#), [HUMANIZE](#), [TWEB](#), [TCSS](#), [PlosOne](#), [JSS](#), [TKDE](#), [IR Journal](#), [SNAM](#), [AJSE](#), [TLT](#), [Computers & Education](#), [Systems & Software](#).
- since 2015 **Program Committee (workshops)**, [BIAS](#), [IronGraphs](#), [MURS](#), [PsiIAS](#), [HAAPIE](#), [MORS](#), [INTRS](#), [Perspectives](#), [RDSM](#), [SOAPS](#), [AFEL](#), [SNAMS](#), [MSM](#), [SynIRgy](#).
- since 2014 **Program Committee (conferences)**, [SIGIR](#), [ECAI](#), [ECIR](#), [ICWE](#), [CIKM](#), [WWW](#), [IUI](#), [HT](#), [EuroCSS](#), [RecSys](#), [WebSci](#), [OpenSym](#), [UMAP](#), [ECTEL](#).
- since 2013 **Open Science & Open-Source Software**, e.g., [TagRec](#), [FairRecSys](#), [aTrain](#).

Membership in International Professional & Academic Associations

- since 2026 **SmartRegulation**, *Smart Regulation field of excellence at University of Graz*, Role: full member.
- since 2025 **GraML**, *Graz Center for Machine Learning at TU Graz*, Role: center board member for core research area on trust.
- since 2024 **Austrian Standards**, *Committee for AI standardization in Austria*, Role: committee member representing Know Center together with Dr. Kerstin Waxnegger.
- since 2023 **Big Data Value Association Task Force**, *Ethical and Trustworthy Artificial and Machine Intelligence (ETAMI) task force of the Big Data Value Association (BDVA)*, Role: member.
- since 2023 **Know Center**, *Works Council of FFG COMET Research Center for Trustworthy AI Know Center*, Role: substitute member (since 2026: full member).
- since 2021 **Frontiers in Big Data**, *Editorial board of Recommender Systems Section*, Role: review editor and research topic editor.
- since 2014 **ACM**, *Association for Computing Machinery (ACM)*, Role: member (since 2017: ACM professional member; since 2026: *ACM IUI community* member in addition).

Media Reactions & Science Communication to the Public

- 2025 **5min.at**, *Grazer Forscher untersuchen wie fair ist KI wirklich?*, News Article on Fairness in AI, *Language*: German.
- 2025 **TU Graz**, *Fairness in AI: Study Shows Central Role of Human Decision-Making*, News Article on Fairness in Multistakeholder Recommender Systems, *Language*: English.
- 2025 **HTS Highlights**, *Trustworthy AI for the Medicine of the Future*, News Article on AI in healthcare in the highlights of the Humantechnology Styria hub, *Language*: English.
- 2024 **SGS Fireside Chat**, *Trustworthy AI: Transparency and Explainability*, Fireside chat as part of the SGS Trustworthy AI series (> 1,000 listeners), *Language*: English.
- 2024 **Trend**, *Der AI Act - Chance für Europa*, News Article on legal aspects of trustworthy AI, *Language*: German.
- 2024 **MediaFutures**, *Transparency, privacy and fairness in recommender systems*, News article on trustworthiness aspects of recommender systems, *Language*: English.
- 2023 **DerStandard**, *Wie Recruiting mit KI in Zukunft auch ohne Diskriminierung möglich ist*, News article on fair AI in the labor market, *Language*: German.
- 2022 **APA Science**, *Digitale Stereotype*, News article on fairness in AI and how digital stereotypes could lead to unintended biases, *Language*: German.
- 2022 **APA Science**, *Wie digitale Stereotype aus der Welt geschafft werden sollen*, News article on popularity bias in recommender systems, *Language*: German.
- 2022 **TU Graz**, *Research and Teaching for Equal Opportunities*, News article about the winners of the TU Graz Mind-the-gap gender and diversity award, *Language*: English.
- 2021 **BiomedCentral**, *Algorithm-Generated Music Recommendations: Low Accuracy for Fans of Beyond-Mainstream Music*, News article on beyond-mainstream users in music recommender systems, *Language*: English.
- 2021 **BackstagePro**, *Metal- und Hip-Hop-Fans haben bei Empfehlungs-Algorithmen das Nachsehen*, News article on niche music recommendations, *Language*: German.
- 2021 **UIBK**, *Auch Musik Algorithmen tun sich manchmal schwer*, News article on music recommender systems, *Language*: German.

Gender Mainstreaming & Leadership Qualifications

- 2025-2026 **Gender & Diversity Training**, [TU Graz Gender, Diversity & Equity unit](#), gender and diversity training for researchers.
- 2025 **Gender & Diversity Award**, [TU Graz Gender, Diversity and Equal Opportunity](#), Awarded for research on fair recommender systems in tourism.
- 2023 **Leadership**, [Wolfgang Eder HR development](#), [Future team lead certificate](#), Know Center Research GmbH team lead programme on gender- and diversity-sensitive leadership.
- 2022-2025 **FFG Femtech**, [Key researcher for FFG Femtech project including 2 Femtech internships](#), gender- and diversity-sensitive recommender systems in Radreisen4All project.
- 2022 **Gender & Diversity Award**, [TU Graz Gender, Diversity and Equal Opportunity](#), Awarded for my research on fairness, diversity, and gender and popularity bias in recommender systems.
- 2023 **Gender & Diversity-sensitive Teaching**, [TU Graz Teaching Academy](#), [Advanced teaching certificate](#), Didactics and communication skills for gender- and diversity-sensitive teaching and mentoring techniques.

Research Talks at International Conferences & Events

- 2026 **ELLIS**, *Invited talk on recommendations and RAG in the digital at the [ELLIS Unit Berlin](#)*, Berlin, Germany.
- 2026 **UMAP**, *Contributing talks on fairness at [UMAP](#)*, Gothenburg, Sweden.
- 2026 **UniGraz**, *Talk on trustworthy AI in the humanities at [University of Graz](#)*, Graz, Austria.
- 2026 **DigitalHumanities**, *Invited talk on recommendations and RAG in the digital humanities breakfast lecture series at [University of Graz](#)*, Graz, Austria.
- 2025 **SmartRegulation**, *Invited panelist on AI regulation at [SmartRegulation](#) field of excellence of University of Graz*, Graz, Austria.
- 2025 **RecSys**, *Contributing talks on fairness at [RecSys](#)*, Prague, Czech Republic.
- 2025 **EWAF**, *Contributing talk on AI certification at [EWAF](#)*, Eindhoven, The Netherlands.
- 2025 **HabilTalk**, *Invited talk on transparency, privacy, and fairness in recommender systems as part of the HabilTalk series of [TU Graz](#)*, Graz, Austria.
- 2024 **RecSys**, *Contributing talk on popularity bias simulation studies at [RecSys](#)*, Bari, Italy.
- 2024 **OEGGF**, *Contributing talk on fair AI in the labor market at the [10. Tagung der Oesterreichischen Gesellschaft für Geschlechterforschung \(OEGGF\)](#)*, Graz, Austria.
- 2024 **AI-Know**, *Contributing talk on evaluation and certification of trustworthy AI at [AI-Know](#)*, Graz, Austria.
- 2024 **MediaFutures**, *Invited talk on trustworthy recommender systems and research/industry collaborations at [MediaFutures SFI](#)*, Bergen, Norway.
- 2024 **Wissenschaftsforum**, *Invited speaker and panelist on fair AI in the labor market at the [Wissenschaftsforum](#)*, Cologne, Germany.
- 2023 **ECIR**, *Contributing talk on recommendations-as-a-service at [ECIR](#)*, Dublin, Ireland.
- 2023 **BIAS**, *Contributing talk on recommendation calibration at [BIAS@ECIR](#)*, Dublin, Ireland.
- 2022 **EBDVA**, *Invited panelist and speaker on trustworthy AI and EU AI Act as part of the [European Big Data Value Forum](#)*, Prague, Czech Republic.
- 2022 **ECIR**, *Contributing talk on news recommendation at [ECIR](#)*, Stavanger, Norway.
- 2022 **BIAS**, *Contributing talk on fairness at [BIAS@ECIR](#)*, Stavanger, Norway.

- 2021 **DataWeek**, *Invited panelist and speaker on breaking silos in data innovation in Europe as part of [BDVA Data Week](#)*, online due to COVID-19.
- 2021 **Science4Future**, *Contributing talk on fairness at [TU Graz](#)*, Graz, Austria.
- 2020 **ECIR**, *Contributing talk on music recommendations at [ECIR](#)*, online due to COVID-19.
- 2019 **REVEAL**, *Contributing talk on recommender systems in data markets at [REVEAL@RecSys](#)*, Copenhagen, Denmark.
- 2019 **EUROCSS**, *Contributing talk on polarization at [EUROCSS](#)*, Zurich, Switzerland.
- 2018 **CIKM**, *Contributing talk on social recommendations at [CIKM](#)*, Turin, Italy.
- 2018 **WebConf**, *Contributing talk on tag recommendations at [WebConf](#)*, Lyon, France.
- 2018 **EUROCSS**, *Contributing talk on confirmation bias in hashtag recommendations [EUROCSS](#)*, Cologne, Germany.
- 2017 **UMAP**, *Contributing talk on tag recommendation at [UMAP](#)*, Bratislava, Slovakia.
- 2017 **WebConf**, *Contributing talk on hashtag extraction at the [WebConf](#)*, Perth, Australia.
- 2017 **UMAP**, *Contributing talk on tag recommendation [UMAP](#)*, Bratislava, Slovakia.
- 2017 **EUROCSS**, *Contributing talk on imbalances in social tagging systems and recommendations [EUROCSS](#)*, London, England.
- 2016 **Hypertext**, *Contributing talk on cognitive modeling at [Hypertext](#)*, Halifax, Canada.
- 2015 **RecSys**, *Contributing talk on recommendation evaluation at [RecSys](#)*, Vienna, Austria.
- 2015 **WWW**, *Contributing talk on recommendations in social tagging systems at [WWW](#)*, Florence, Italy.
- 2015 **CSSWS**, *Contributing talk on tag recommendations at [CSSWS](#)*, Cologne, Germany.
- 2015 **i-KNOW**, *Contributing talk on tourism recommender systems at [i-KNOW](#)*, Graz, Austria.
- 2014 **WWW**, *Contributing talk on cognitive modeling at [WWW](#)*, Seoul, Korea.
- 2013 **i-Semantics**, *Contributing talk on social semantic services at [i-Semantics](#)*, Graz, Austria.

International Research Projects

- 2026–2029 **HorizonEurope**, [REPLICATE](#), Led by [University of Coimbra](#), Role: study co-lead for AI replications and open science.
- 2026–2029 **HorizonEurope**, [EU MARitime SEcurity through Autonomous Real-time Surveillance platforms \(MARSEC\)](#), Led by [Idener.ai](#), Role: task lead on AI ethics.
- 2026–2029 **HorizonEurope**, [CompreHensive EVALuation methods for GPAI models \(CHEVAL\)](#), Led by [FORTH](#), Role: co-work package lead on LLM evaluation.
- 2026–2029 **HorizonEurope**, [Navigational AI assistant for Virtual Worlds Immersion \(NAVI\)](#), Led by [Idener.ai](#), Role: co-work package lead on LLMs.
- 2026–2029 **FFG COMET Module**, [Interfaces of Agent-Centric AI \(IACAI\)](#), Led by [Know Center](#), with partners, e.g., [TU Munich](#), Role: key researcher for subproject on societal aspects.
- 2026–2028 **Erasmus+**, [AIDEA](#), Led by [CIAPE](#), Role: key researcher for AI methods in design.
- 2026–2027 **FFG BASIS**, [CareLLM](#), Led by [DaphOS](#), Role: key researcher for trustworthy LLMs.
- 2025–2027 **HorizonEurope**, [Linked User-driven Multidisciplinary Exploration Network \(LUMEN\)](#), Led by [CNRS](#), Role: key researcher for content-based recommendation systems.
- 2024–2026 **HorizonEurope**, [Enhancing Trust, Integrity and Efficiency in Research through next-level Reproducibility \(TIER-2\)](#), Led by [Know Center Research GmbH](#), Role: key researcher.

- 2024–2025 **FFG BASIS**, *Prognosen von Ressourcenbedarfen (Pro'K'ress)*, Led by [DaphOS](#), Role: key researcher on privacy-aware machine learning.
- 2024–2025 **OpenWebSearch Third-Party Tender**, *TILDE*, Led by [University of Passau](#), Role: key researcher on bias mitigation in LLMs.
- 2024–2025 **FFG AI4Green**, *Strategic AI Roadmap for Mobility (SAIROM)*, Led by [University of Graz](#), Role: key researcher on sustainability aspects.
- 2024–2025 **Styrian AI Future Fund**, *FairRecSys*, Led by [TU Graz](#), Role: work package lead on multistakeholder AI aspects.
- 2023–2026 **FFG COMET Research Center**, *Know Center - Research Center for Trustworthy AI*, Led by [Know Center Research GmbH](#), COMET-funded research projects conducted with e.g., [SGS](#), Role: research area head for FAIR-AI.
- 2022–2026 **FFG COMET Module**, *Data-Driven Immersive Analytics (DDIA)*, Led by [Know Center Research GmbH](#), research conducted with partners, e.g., [University of Stuttgart](#), Role: key researcher for subproject on recommendations.
- 2022–2025 **FFG FemTech**, *Radreisen4All*, Led by [CycleBee](#), Role: key researcher in the field of multistakeholder recommender systems.
- 2020–2023 **FFG COMET Module**, *Data-Driven Artificial Intelligence (DDAI)*, Led by [Know Center Research GmbH](#), research conducted with partners, e.g., [TU Delft](#), Role: key researcher for subproject on XAI and private AI.
- 2020–2023 **Erasmus+**, *Cogsteps*, Led by [University of Zagreb](#), Role: key researcher on recommender systems for matchmaking.
- 2020–2022 **Horizon2020**, *Trusted Secure Data Sharing Space (TRUSTS)*, Led by [Leibniz University Hannover](#), Role: task lead on AI-based matchmaking.
- 2020–2022 **Horizon2020**, *TRIPLE*, Led by [CNRS](#), Role: co-task lead on scholarly recommender systems.
- 2020–2022 **Horizon2020**, *AI4EU*, Led by [Thales SIX GTS France SAS](#), Role: co-task lead on European data for AI.
- 2019–2022 **FFG COMET Research Center**, *Know Center - Research Center for Big Data Analytics*, Led by [Know Center Research GmbH](#), COMET-funded research projects conducted with e.g., [DiePresse](#), Role: deputy research area head.
- 2019–2021 **FFG BASIS**, *Automated Marketing and Loyalty System for Retailers and Stores (Joloo)*, Led by [Joloo GmbH](#), Role: researcher for recommender systems.
- 2018–2020 **FFG BASIS**, *Studo App Sales Offensive*, Led by [Moshbit GmbH](#), Role: researcher for recommender systems.
- 2018–2020 **Styrian Health Fund**, *Health-Literacy und Diversity (HeLi-D)*, Led by [University of Graz](#), Role: work package lead on personalization.
- 2018–2020 **OpenAIRE Third-Party Tender**, *OpenAIRE Matchmaker*, Led by [OpenAire](#), Role: researcher for matchmaking algorithms.
- 2015–2018 **Horizon2020**, *Analytics for Everyday Learning (AFEL)*, Led by [University of Galway](#), Role: co-task lead on scholarly recommender systems.
- 2015–2018 **Horizon2020**, *MoreGrasp*, Led by [TU Graz](#), Role: researcher for recommender systems.
- 2015–2018 **FFG Lighthouse**, *Data Market Austria (DMA)*, Led by [RSA](#), Role: researcher for recommender systems.
- 2012–2016 **FP7**, *Learning Layers*, Led by [Tallinn University](#), Role: Ph.D. candidate and co-work package lead.

- 2012–2014 **FP7**, *Organic Lingua*, Led by [University of Alcalá](#), Role: researcher.
 2010–2012 **FP7**, *ALICE*, Led by [University of Milano-Bicocca](#), Role: researcher.

Academic Self-Administration & Institutional Service

- since 2026 **Habilitation Committee Member**, [University of Graz](#), Faculty of Humanities, in my role as member of the professorial curia and representing Digital Humanities.
- since 2026 **Works Council Member**, [Know Center](#), “Betriebsratsmitglied”, elected representative for employee interests, participating in corporate governance & mediation.
- since 2026 **Tenure-Track Professorship Hiring Committee Member**, [University of Graz](#), Computational Humanities, Department of Digital Humanities.
- since 2026 **External Expert for Professorship Promotions**, [University of Gothenburg](#), Sweden, international expert evaluator for external Full Professor promotion committees.
- 2026 **Ph.D. Thesis Examination Committee Member**, [University of Graz](#), Lucija Krusic: *NLP in Digital Humanities*, Ph.D. exam with Prof. Eva Pfanzelter, [University of Innsbruck](#); Prof. Georg Vogeler, [University of Graz](#); and Chiara Zuanni, [University of Continuing Education Krems](#).
- 2026 **Master’s Thesis Examination Committee Member**, [University of Graz](#), Lukas Waldhofer: *Data Analysis Tool for Humanities Scholars*, with Prof. Georg Vogeler representing [University of Graz](#).
- 2025–2026 **External Master’s Thesis Supervisor**, [University of Vienna](#), Anja Rejc: *Cognitive Biases and Trustworthy AI*, served as external academic advisor and evaluator.
- since 2025 **External Expert for Faculty Appointments**, [University of Bergen](#), Norway, International expert evaluator for competitive academic faculty hires..
- 2025 **Master’s Thesis Examination Committee Chairperson**, [TU Graz](#), Thomas Zenkl: *The Trouble with Algorithms*, with Assoc.Prof. Bernhard Wieser representing [TU Graz](#).
- 2024 **Master’s Thesis Examination Committee Chairperson**, [TU Graz](#), Sara Skardelly: *Environmental (In)justice*, with Assoc.Prof. Bernhard Wieser representing [TU Graz](#).

Students Supervised & Mentored

- 2026 **Master’s thesis**, [Norwegian University of Science and Technology](#), Kaia Kolstad: *Fairness Trade-Offs in Multistakeholder Recommender Systems*, External co-supervision with Assoc.Prof. Özlem Özgöbek, [Norwegian University of Science and Technology](#).
- 2026 **Bachelor thesis**, [TU Graz](#), Margarita Dubina: *LLMs and RAG in Digital Humanities*.
- 2026 **Master’s thesis**, [TU Graz](#), Michael Wittmayer: *LLM-based Research Assistants*.
- since 2026 **Ph.D. thesis**, [TU Graz](#), Andrea Forster: *Multistakeholder Aspects of Agentic AI*.
- since 2026 **Ph.D. thesis**, [TU Graz](#), Jenish Thapa: *Explainability in Multi-Agent AI Systems*, Co-supervision with Assoc. Prof. Viktoria Pammer-Schindler, [TU Graz](#).
- since 2026 **Master’s thesis**, [University of Graz](#), Ulrich Schütz: *RAG and LLMs in Digital Humanities*, Co-supervision with Prof. Joelle Weis, [Trier Center for Digital Humanities](#).
- since 2026 **Master’s thesis**, [University of Graz](#), Chiara Citro: *Dataset Biases in Digital Humanities*, Co-supervision with Prof. Chiara Zuanni, [University of Continuing Education Krems](#).
- 2025–2026 **Master’s thesis**, [TU Graz](#), Fanni Maria Tuominen: *Computational Antitrust*, Co-supervision with Dr. Jürgen Fleiß, [University of Graz](#).
- 2025–2026 **Master’s thesis**, [University of Vienna](#), Anja Rejc: *Cognitive Biases and Trustworthy AI*, External supervisor.

- 2025-2026 **Master's thesis**, [TU Graz](#), Juergen Stumpf: *Regulation and Governance of AutoML Tools in Industry*, Co-supervision with Prof. Stefan Thalmann, [University of Graz](#).
- 2025 **Master's thesis**, [TU Graz](#), Valentin Forster: *Detecting Price Anomalies Indicative of Antitrust Violations with AI*, Co-supervision with Dr. Juergen Fleiß, [University of Graz](#).
- since 2024 **Ph.D. thesis**, [TU Graz](#), Florian Atzenhofer-Baumgartner: *Recommender Systems in Digital Humanities*.
- 2024-2025 **Master's thesis**, [TU Graz](#), Andrea Forster: *Popularity Bias in Point-of-interest Recommender Systems*, Co-supervision with Prof. Stefan Thalmann, [University of Graz](#).
- 2024 **Internship**, [Know Center Research GmbH](#), Adrian Marangoni: *Fair AI in Mobility: Literature Review and Policy Recommendations*.
- 2024 **Master's thesis and internship**, [TU Graz](#), Ioana Serban: *Bias Mitigation in Public Machine Learning Datasets*.
- 2024 **Bachelor thesis**, [TU Graz](#), Gregor Autischer: *Practical Aspects of AI Certification in the Light of the EU AI Act*.
- 2023 **Bachelor thesis**, [TU Graz](#), Harald Semmelrock: *Reproducibility in AI-driven Research*.
- 2023 **Bachelor thesis**, [TU Graz](#), Michael Pöchlinger: *Data Quality and Biases in Public Machine Learning Training Datasets*.
- 2023 **Internship**, [Know Center Research GmbH](#), Gökay Yildirim: *Popularity Bias Detection*.
- 2022 **Bachelor thesis and Master's project**, [TU Graz](#), Gregor Mayr: *Calibration in Recommender Systems*, Co-supervision with: Prof. Elisabeth Lex, [TU Graz](#).
- 2020-2026 **Ph.D. thesis**, [TU Graz](#), Peter Müllner: *Privacy in Recommender Systems*, Co-supervision with: Prof. Elisabeth Lex, [TU Graz](#).
- 2020-2026 **Ph.D. thesis**, [TU Graz](#), Tomislav Duricic: *Sparsity and Interpretability of Graph-based Recommender Systems*, Co-supervision with: Prof. Elisabeth Lex, [TU Graz](#).
- 2020 **Master's thesis**, [TU Graz](#), Mario Wagner: *Diversity-Aware Recommendation of Tweets*, Co-supervision with: Prof. Elisabeth Lex, [TU Graz](#).
- 2019 **Master's thesis**, [TU Graz](#), Peter Müllner: *Studying Non-Mainstream Listening Behavior For Music Recommendations*, Co-supervision with: Prof. Elisabeth Lex, [TU Graz](#).
- 2016 **Bachelor thesis**, [TU Graz](#), Andreas Punz: *Detection and Analysis of Communities on Twitter*, Co-supervision with: Prof. Elisabeth Lex, [TU Graz](#).

University Courses Taught

- since 2026 **Privatissimum: Theory and Application of Digital Humanities**, [University of Graz](#), 4 ECTS, Master Digital Humanities, Doctoral Programme in Philosophy at Faculty of Arts and Humanities.
Role: [providing an advanced research seminar](#) for Master and Ph.D. students (~15 students)
- since 2026 **Project Seminar**, [University of Graz](#), 6 ECTS, Master Digital Humanities.
Role: [supervision](#) of practical student projects applying (trustworthy) AI-driven methods to historical datasets and digital editions (~ 15 students)
- since 2025 **Introduction to Computer Science**, [University of Graz](#), 5 ECTS, Master Digital Humanities.
Role: [lecturer](#) (~30 students), restructured course to include modern computer science aspects
- since 2024 **Data Management**, [TU Graz](#), 4 ECTS, Bachelor Software Engineering & Management, Computer Science, Information & Computer Engineering.
Role: [lecturer](#) (~500 students), restructured course focusing on practical aspects of data management and databases

- since 2024 **Foundations of Digitalization**, [University of Graz](#), 4 ECTS, Bachelor Business Administration, Economics, Sociology.
Role: [lecturer responsible for course content on relational databases](#) (~500 students)
- since 2024 **Privatissimum: Trustworthy AI and recommender systems**, [TU Graz](#), Doctoral school Computer Science.
Role: [mentoring Ph.D. students](#) (~1 to 2 students)
- 2024 **Case Studies Business Analytics**, [University of Graz](#), Guest lecture, AI reproducibility.
Role: [guest lecturer on reproducibility of AI-driven research](#) (~5 students)
- since 2023 **Databases**, [TU Graz](#), 3 ECTS, Bachelor Information & Computer Engineering, Master Computational Social Systems, Electrical & Audio Engineering.
Role: [lecturer](#) (~100 students), restructured course using a project-based teaching concept for students without a computer science background
- since 2023 **Introduction to Scientific Writing**, [TU Graz](#), 2 ECTS, Bachelor Information & Computer Engineering, Computer Science, Software Engineering & Management.
Role: [offering seminar topics on trustworthy AI and recommender systems](#) (~1 to 10 students)
- 2022 **Journalism & Public Relations**, [FH Joanneum](#), Guest lecture, Recommender systems.
Role: [guest lecturer on news recommender systems for journalists](#) (~50 students)
- 2016 **Science 2.0**, [TU Graz](#), 3 ECTS, Master Software Engineering & Management, Computer science.
Role: [responsible for the exercises](#) (~50 students), *main lecturer*: Prof. Elisabeth Lex
- 2014 **Recommender Systems**, [PUC Chile](#), Practical assignment, Content-based recommender systems.
Role: [guest course assistant](#) (~50 students), *main lecturer*: Prof. Denis Parra ([PUC Chile](#))
- 2014 **Web Science & Web Technology**, [TU Graz](#), 3 ECTS, Bachelor Software Engineering & Management, Master Information & Computer Engineering.
Role: [responsible for the exercises](#) (~200 students), *main lecturer*: Prof. Christoph Trattner
- 2011–2012 **Information Search & Retrieval**, [TU Graz](#), 5 ECTS, Master Software Engineering & Management, Computer Science, Information & Computer Engineering.
Role: [responsible for the exercises](#) (~100 students), *main lecturers*: Assoc.Prof. Christian Gütl, Prof. Mohammad Al-Smadi
- 2009 **Data Structures & Algorithms**, [TU Graz](#), 1.5 ECTS, Bachelor Biomedical Engineering, Technical Mathematics, Software Engineering & Management, Computer Science, Information & Computer Engineering.
Role: [responsible for one exercise group](#) (~30 students), *main lecturer*: Prof. Robert Legenstein

International Academic References

- Prof. Robin Burke **Full Professor**, [University of Colorado Boulder](#), USA.
Email: robin.burke@colorado.edu | **Web**: [Link](#)
Connection: Scientific advisor of my FAIR-AI group at Know Center; co-author on multistakeholder fairness in trustworthy AI and recommender systems in digital humanities.
- Prof. Maria Soledad Pera **Associate Professor**, [TU Delft](#), The Netherlands.
Email: m.s.pera@tudelft.nl | **Web**: [Link](#)
Connection: Co-author on algorithmic fairness and bias mitigation in information retrieval and machine learning approaches for marginalized user groups, especially children.
- Prof. Christoph Trattner **Full Professor & Director of MediaFutures**, [University of Bergen](#), Norway.
Email: christoph.trattner@uib.no | **Web**: [Link](#)
Connection: Ph.D. supervisor in the first two years of my Ph.D. studies; long-term collaborator on human-centric AI, machine learning, and psychology-informed recommender systems.

Univ.-Prof. Dr. Dominik Kowald

List of Publications (* indicates equal contributions)

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ORCID: <https://orcid.org/0000-0003-3230-6234>

Website: <https://dominikkowald.info>

Monographs

- [1] **Dominik Kowald**. Transparency, privacy, and fairness in recommender systems. *Habilitation thesis in Applied Computer Science at TU Graz*, 2024. URL <https://doi.org/10.48550/arXiv.2406.11323>.
- [2] **Dominik Kowald**. *Modeling activation processes in human memory for tag recommendations*. OMNI Scriptum Publishing, 2017. ISBN 978-6202320726. URL https://dominikkowald.info/documents/2017dissertation_book.pdf.
- [3] **Dominik Kowald**. Modeling activation processes in human memory to improve tag recommendations. *Ph.D. thesis at TU Graz*, 2017. URL <https://doi.org/10.48550/arXiv.1803.03176>.

(Co-)edited Volumes and Peer-Reviewed Conference Contributions

- [4] Florian Atzenhofer-Baumgartner and **Dominik Kowald**. What do humanities scholars need? A user model for recommendation in digital archives. In *UMAP'26*, 2026. URL <https://doi.org/10.1145/3774935.3806171>.
- [5] Robin Ungruh, Alejandro Bellogín, **Dominik Kowald**, and Maria Soledad Pera. When preference is not enough: Why recommender systems require human-aware evaluation for children. In *UMAP'26*, 2026. URL <https://doi.org/10.1145/3774935.3812719>.
- [6] Peter Müllner, **Dominik Kowald**, Markus Schedl, and Elisabeth Lex. Meta-learning and targeted differential privacy to improve the accuracy–privacy trade-off in recommendations. In *UMAP'26*, 2026. URL <https://doi.org/10.1145/3774935.3812709>.
- [7] Andrea Forster, Peter Müllner, Denis Helic, Elisabeth Lex, and **Dominik Kowald**. Fair agents: Balancing multistakeholder alignment in multi-agent personalization systems. In *LLM4Good@UMAP'26*, 2026. URL <https://ceur-ws.org/Vol-4206/LLM4Good-3.pdf>.
- [8] Gregor Autischer, Kerstin Waxnegger, and **Dominik Kowald**. Towards EU AI Act compliance: Self-certification and fairness alignment for facial emotion recognition. In *ECAF'26*, 2026. URL <https://ssrn.com/abstract=7039959>.
- [9] Andrea Forster, Gregor Autischer, **Dominik Kowald**, and Simone Kopeinik. From assessment to aggregation: A multistakeholder framework for identifying and mitigating biases in AI-driven job profiling and recommendation. In *BIAS'2026: Mitigating Diversity Biases of AI in the Labour Market*, 2026. URL <https://doi.org/10.48550/arXiv.2607.15707>.
- [10] Anja Rejc, Juliana Madritsch, Neven Elsayed, **Dominik Kowald**, and Simone Kopeinik. An investigation of trust and overreliance on conversational instructions in augmented reality. In *Conversational User Interfaces (CUI'26)*, 2026. URL <https://doi.org/10.1145/3816046.3816296>.
- [11] Florian Atzenhofer-Baumgartner, Georg Vogeler, and **Dominik Kowald**. A multistakeholder approach to value-driven co-design of recommender system evaluation metrics in digital archives. In *RecSys'25*, 2025. URL <https://doi.org/10.1145/3705328.3748026>.

- [12] Robin Ungruh, Alejandro Bellogín, **Dominik Kowald**, and Maria Soledad Pera. Impacts of mainstream-driven algorithms on recommendations for children across domains: A reproducibility study. In *RecSys'25*, 2025. URL <https://doi.org/10.1145/3705328.3748160>.
- [13] Andrea Forster, Simone Kopeinik, Denis Helic, Stefan Thalmann, and **Dominik Kowald**. Exploring the effect of context-awareness and popularity calibration on popularity bias in POI recommendations. In *RecSys'25*, 2025. URL <https://doi.org/10.1145/3705328.3748017>.
- [14] **Dominik Kowald**. Investigating popularity bias amplification in recommender systems employed in the entertainment domain. In *EWAF'25*, 2025. URL <https://proceedings.mlr.press/v294/kowald25a.html>.
- [15] Gregor Autischer, Kerstin Waxnegger, and **Dominik Kowald**. AI certification and assessment catalogues: Practical use and challenges in the context of the European AI Act. In *EWAF'25*, 2025. URL <https://proceedings.mlr.press/v294/autischer25a.html>.
- [16] Elisabeth Lex, Kevin Innerebner, Marko Tkalcic, **Dominik Kowald**, and Markus Schedl. Hybrid AI for human-centric personalization (HyPer). In *UMAP'25 Adjunct*, 2025. URL <https://doi.org/10.1145/3708319.3727563>.
- [17] Kevin Innerebner, **Dominik Kowald**, Markus Schedl, and Elisabeth Lex. Hybrid personalization using declarative and procedural memory modules of the cognitive architecture ACT-R. In *UMAP'25 Adjunct*, 2025. URL <https://doi.org/10.1145/3708319.3734176>.
- [18] Oleg Lesota, Jonas Geiger, Max Walder, **Dominik Kowald**, and Markus Schedl. Oh, behave! Country representation dynamics created by feedback loops in music recommender systems. In *RecSys'24*, 2024. URL <https://doi.org/10.1145/3640457.3688187>.
- [19] Tomislav Duricic, Peter Müllner, Nicole Weidinger, Neven Elsayed, **Dominik Kowald**, and Eduardo Veas. AI-powered immersive assistance for interactive task execution in industrial environments. In *ECAI'24*, 2024. URL <https://www.doi.org/10.3233/FAIA241037>.
- [20] Gustavo Escobeda, Marta Moscati, Peter Müllner, Simone Kopeinik, **Dominik Kowald**, Elisabeth Lex, and Markus Schedl. Making Alice appear like Bob: A probabilistic preference obfuscation method for implicit feedback recommendation models. In *ECML-PKDD'24*, 2024. URL https://doi.org/10.1007/978-3-031-70368-3_21.
- [21] Peter Müllner, Elisabeth Lex, Markus Schedl, and **Dominik Kowald**. The impact of differential privacy on recommendation accuracy and popularity bias. In *ECIR'24*, 2024. URL https://doi.org/10.1007/978-3-031-56066-8_33.
- [22] Florian Königsdorfer, Armin Haberl, **Dominik Kowald**, Tony Ross-Hellauer, and Stefan Thalmann. Black box or open science? A study on reproducibility in AI development papers. In *HICSS'24*, 2024. URL <https://hdl.handle.net/10125/106458>.
- [23] **Dominik Kowald**, Deqing Yang, and Emanuel Lacic, editors. *Reviews in recommender systems*, 2024. Frontiers Media. URL <https://doi.org/10.3389/978-2-8325-4766-3>.
- [24] Florian Atzenhofer-Baumgartner, Bernhard Geiger, Georg Vogeler, and **Dominik Kowald**. Value identification in multistakeholder recommender systems for humanities and historical research: The case of the digital archive Monasterium.net. In *Normalize@RecSys'24*, 2024. URL <https://ceur-ws.org/Vol-3898/paper4.pdf>.
- [25] Florian Atzenhofer-Baumgartner, Bernhard Geiger, Christoph Trattner, Georg Vogeler, and **Dominik Kowald**. Challenges in implementing a recommender system for historical research in the humanities. In *AltRecSys@RecSys'24*. ACM, 2024. URL <https://doi.org/10.48550/arXiv.2410.20909>.

- [26] Marta Moscati, Christian Wallmann, Markus Reiter-Haas, **Dominik Kowald**, Elisabeth Lex, and Markus Schedl. Integrating the ACT-R framework with collaborative filtering for explainable sequential music recommendation. In *RecSys'23*, 2023. URL <https://doi.org/10.1145/3604915.3608838>.
- [27] Emanuel Lacic, Tomislav Duricic, Leon Fadljevic, Dieter Theiler, and **Dominik Kowald**. Uptrendz: API-centric real-time recommendations in multi-domain settings. In *ECIR'23*, 2023. URL https://doi.org/10.1007/978-3-031-28241-6_23.
- [28] **Dominik Kowald***, Gregor Mayr*, Markus Schedl, and Elisabeth Lex. A study on accuracy, miscalibration, and popularity bias in recommendations. In *BIAS'23*, pages 1–16, 2023. URL https://doi.org/10.1007/978-3-031-37249-0_1.
- [29] Peter Muellner, Stefan Schmerda, Dieter Theiler, Stefanie Lindstaedt, and **Dominik Kowald**. Towards employing recommender systems for supporting data and algorithm sharing. In *DataEconomy@CoNext'22*, 2022. URL <https://doi.org/10.1145/3565011.3569055>.
- [30] Emanuel Lacic, Leon Fadljevic, Franz Weissenboeck, Stefanie Lindstaedt, and **Dominik Kowald**. What drives readership? An online study on user interface types and popularity bias mitigation in news article recommendations. In *ECIR'22*, 2022. URL https://doi.org/10.1007/978-3-030-99739-7_20.
- [31] **Dominik Kowald** and Emanuel Lacic. Popularity bias in collaborative filtering-based multimedia recommender systems. In *BIAS'22*, 2022. URL https://doi.org/10.1007/978-3-031-09316-6_1.
- [32] Emanuel Lacic and **Dominik Kowald**. Recommendations in a multi-domain setting: Adapting for customization, scalability and real-time performance. In *Industry-Day Track of ECIR'22*, 2022. URL <https://doi.org/10.48550/arXiv.2203.01256>.
- [33] Peter Muellner, **Dominik Kowald**, and Elisabeth Lex. Robustness of meta matrix factorization against strict privacy constraints. In *ECIR'21*, 2021. URL <https://doi.org/10.1007/978-3-030-72240-1>.
- [34] Peter Müllner, Elisabeth Lex, and **Dominik Kowald**. Impact of meta learning for privacy-preserving recommender systems. In *TRAIF'21*, 2021. URL <https://doi.org/10.5281/zenodo.15221745>.
- [35] Oleg Lesota, Alessandro Melchiorre, Navid Rekabsaz, Stefan Brandl, **Dominik Kowald**, Elisabeth Lex, and Markus Schedl. Analyzing item popularity bias of music recommender systems: Are different genders equally affected? In *RecSys'21*, 2021. URL <https://doi.org/10.1145/3460231.3478843>.
- [36] Tomislav Duricic, **Dominik Kowald**, Markus Schedl, and Elisabeth Lex. My friends also prefer diverse music: Homophily and link prediction with user preferences for mainstream, novelty, and diversity in music. In *MSDNS@ASONAM'21*, 2021. URL <https://doi.org/10.1145/3487351.3492706>.
- [37] Peter Muellner, Elisabeth Lex, and **Dominik Kowald**. Position paper on simulating privacy dynamics in recommender systems. In *SimuRec@RecSys'21*, 2021. URL <https://doi.org/10.48550/arXiv.2109.06473>.
- [38] **Dominik Kowald**, Markus Schedl, and Elisabeth Lex. The unfairness of popularity bias in music recommendation: A reproducibility study. In *ECIR'20*, pages 35–42, 2020. URL https://doi.org/10.1007/978-3-030-45442-5_5.
- [39] **Dominik Kowald***, Elisabeth Lex*, and Markus Schedl. Utilizing human memory processes to model genre preferences for personalized music recommendations. In *HUMANIZE@IUI'20*. Association of Computing Machinery, 2020. URL <https://doi.org/10.48550/arXiv.2003.10699>.
- [40] Tomislav Duricic, Hussain Hussain, Emanuel Lacic, **Dominik Kowald**, Denis Helic, and Elisabeth Lex. Empirical comparison of graph embeddings for trust-based collaborative filtering. In *ISMIS'20*, 2020. URL https://doi.org/10.1007/978-3-030-59491-6_17.

- [41] Leon Fadljevic, Katharina Maitz, **Dominik Kowald**, Viktoria Pammer-Schindler, and Barbara Gasteiger-Klipcera. Slow is good: The effect of diligence on student performance in the case of an adaptive learning system for health literacy. In *LAK'20*, 2020. URL <https://doi.org/10.1145/3375462.3375502>.
- [42] Simone Kopeinik, Elisabeth Lex, **Dominik Kowald**, Dietrich Albert, and Paul Seitlinger. A real-life school study of confirmation bias and polarisation in information behaviour. In *ECTEL'19*, 2019. URL https://doi.org/10.1007/978-3-030-29736-7_31.
- [43] Emanuel Lacic*, **Dominik Kowald***, Dieter Theiler, Matthias Traub, Lucky Kuffer, Stefanie Lindstaedt, and Elisabeth Lex. Evaluating tag recommendations for e-book annotation using a semantic similarity metric. In *REVEAL@RecSys'19*, 2019. URL <https://doi.org/10.48550/arXiv.1908.04042>.
- [44] **Dominik Kowald**, Matthias Traub, Dieter Theiler, Heimo Gursch, Stefanie Lindstaedt, Roman Kern, and Elisabeth Lex. Using the open Meta Kaggle dataset to evaluate tripartite recommendations in data markets. In *REVEAL@RecSys'19*, 2019. URL <https://doi.org/10.48550/arXiv.1908.04017>.
- [45] **Dominik Kowald***, Elisabeth Lex*, and Markus Schedl. Modeling artist preferences for personalized music recommendations. In *ISMIR'19*, 2019. URL <https://archives.ismir.net/ismir2019/latebreaking/000001.pdf>.
- [46] Elisabeth Lex and **Dominik Kowald**. The impact of time on hashtag reuse in twitter: A cognitive-inspired hashtag recommendation approach. In *INFORMATIK'19*, 2019. URL https://doi.org/10.18420/inf2019_46.
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